

Education

Bachelor of Science in Applied Mathematics & Computer Science

University of Delaware, Newark, DE

Honors College | Expected May 2027

Work Experience

- **Data Science Intern — Victory Capital Management (Python, SQL, K-Means Clustering)**
 - Pulled user-level data from Amazon Redshift with SQL to build datasets for K-Means clustering, enabling actionable insights into customer behavior
 - Diagnosed and resolved data and modeling issues in a production K-Means pipeline, including low-signal features, misinterpreted clusters, and inconsistent behavioral definitions
 - Redesigned feature engineering pipeline, increasing silhouette scores and producing more interpretable clusters, helping the marketing team target user segments more effectively
 - Streamlined cluster usability for downstream analytics, improving decision-making for marketing campaigns

Selected Machine Learning Projects

- **Sports Analytic Predictions (Python, XGBoost):**
 - Built an end-to-end XGBoost pipeline on real NFL rushing yard game data, achieving actionable predictive insights for rushing yards and over/under outcomes
 - Engineered rolling workload, efficiency, and matchup features from historical game data
 - Achieved RMSE of 28.87 yards and R^2 of 0.39 on seven seasons of out-of-sample data; classification model achieved ROC-AUC of 0.563 over a two season evaluation period
- **Trading Bots (Python, AWS, XGBoost):**
 - Designed and deployed Python-based algorithmic trading bots on AWS using live WebSocket market data for real-time SPY options execution
 - Integrated XGBoost predictive models to drive live decisions

Technical Skills

- **Languages:** Python, SQL, C++, C, JavaScript, TypeScript
- **Machine Learning / Data Science:** scikit-learn (K-Means, model evaluation), XGBoost, feature engineering, statistical modeling, exploratory data analysis, hypothesis testing
- **Data & Visualization:** Pandas, NumPy, Matplotlib, Seaborn, Plotly
- **Cloud & Infrastructure:** AWS (S3, Redshift, ECS, ECR), Docker, Firebase
- **Developer Tools & Systems:** Git, Bash, Jupyter Notebook, Visual Studio Code, WebSocket-based streaming

University Activities

- **Paid Instructional Assistant – General Computer Science for Engineers**
 - Lead lectures, assisting engineering students new to computer science in understanding core programming concepts
 - Provide one-on-one support to students in Python programming, troubleshooting code, and reinforcing best practices